

Literature status of Open Problems 3–4 in arXiv:math/0101227

Autonomous mathematical discovery run `fa_banach_001`, lane 8

12 August 2026

Status. `literature_already_answered` for Open Problem 3 in the canonical birth–death and half-line diffusion settings; substantial but not exact later-literature progress for Open Question 4.

Original questions

Mu-Fa Chen asks on source PDF page 6:

Open problem 3. Find out a criterion for the Nash inequality in dimension one.

Open question 4. Does there exist a variational formula in the form of part (2) of Theorem 4.1 for the logarithmic Sobolev or Nash inequality?

The source’s Added in Proof excludes these two questions from the problems solved before publication [1].

Problem 3: explicit later answer

Mao’s introduction explicitly says that the paper addresses the question posed in Chen’s source and describes the answer as complete for birth–death processes [2]. For an ergodic birth–death process with invariant probability masses μ_k , death rates a_k , and $\nu > 2$, Theorem 2.2(ii) states that the centered Nash inequality holds if and only if

$$\sup_{n \geq 1} \left(\sum_{k \geq n} \mu_k \right)^{(\nu-2)/\nu} \sum_{k=1}^n \frac{1}{\mu_k a_k} < \infty.$$

This is the source table’s proposed condition after using $\mu_k a_k = \mu_{k-1} b_{k-1}$, with the stated epsilon gap removed.

For the ergodic half-line diffusion $L = a(x)d^2/dx^2 + b(x)d/dx$, put $C(x) = \int_0^x b/a$ and normalize $d\mu = Z^{-1}a^{-1}e^C dx$. Mao’s Theorem 3.1(ii) gives the corresponding if-and-only-if criterion

$$\sup_{x > 0} \left(\int_x^\infty \frac{e^{C(y)}}{a(y)} dy \right)^{(\nu-2)/\nu} \int_0^x e^{-C(y)} dy < \infty.$$

The same paper treats transient versions and explicit two-sided bounds for the optimal constants. The supporting authors therefore knew they were answering the extracted problem; this is not an agent-inferred implication.

Question 4: substantial but scoped later progress

Chen’s 2002 paper extends one-dimensional Poincaré variational methods to a large class of Banach/Orlicz norms, motivated explicitly by Nash and logarithmic-Sobolev inequalities [3]. The resulting theorem gives an exact finiteness criterion, explicit factor bounds, and two inf–sup expressions that are labeled formulas for *upper bounds*. The 2003 birth–death analogue gives the discrete criteria and bounds [4]. Specializing the Orlicz norm yields the Nash and logarithmic-Sobolev criteria.

This is an affirmative answer only under a weakened reading of “variational formula.” The papers do not identify the sharp centered Nash or entropy constant with both a sup–inf and an inf–sup expression, as Theorem 4.1(2) does for the ordinary Poincaré constant. The exact dual equality should therefore not be marked literature-closed on this evidence.

Search evidence and files

The audit used the source paper, Mao’s explicit cross-reference to Chen’s page-335 question, Theorems 2.2 and 3.1 of Mao, the abstracts and theorem summaries of Chen’s 2002/2003 papers, and the later monograph’s Chapter 6 summary (Theorems 6.4 and 6.8–6.12). The local folder includes the original source PDF. Stable supporting DOI links are below; publisher and author-hosted PDF downloads were blocked in this environment.

A separate partial-result packet proves an exact two-site nonlinear dual formula for every $p > 2$ and records why the centered problem remains harder.

References

- [1] M.-F. Chen, *Explicit criteria for several types of ergodicity*, Chinese J. Appl. Probab. Statist. 17 (2001), 1–8; [arXiv:math/0101227](#).
- [2] Y.-H. Mao, *Nash inequalities for Markov processes in dimension one*, Acta Math. Sin. (Engl. Ser.) 18 (2002), 147–156; [doi:10.1007/s101140100128](#).
- [3] M.-F. Chen, *Variational formulas of Poincaré-type inequalities in Banach spaces of functions on the line*, Acta Math. Sin. (Engl. Ser.) 18 (2002), 417–436; [doi:10.1007/s10114-002-0176-8](#).
- [4] M.-F. Chen, *Variational formulas of Poincaré-type inequalities for birth–death processes*, Acta Math. Sin. (Engl. Ser.) 19 (2003), 625–644; [doi:10.1007/s10114-003-0308-9](#).